LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine Health monitoring by using Power performance Check
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine Health monitoring by using Power performance Check
5. Specific Objectives	26.1.1.8 Trainee will be able to explain vibration analysis EHM Technique

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Vibration Analysis EHM data sheets; Instructor supervision on interpretation of amplitude/frequency trends
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engine Health Monitoring?
(b) Define Power Performance Check in the context of EHM.
(c) List three common methods used for vibration analysis.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Vibration Analysis EHM • Sensors and Transducers in Engine Monitoring • Frequency Domain Analysis Fundamentals • Amplitude and Phase Interpretation • Thresholds and Alarm Settings	L & Ds	PPT, IB	15 min
(b)	• Data Acquisition Systems Overview • Signal Processing Techniques • Fault Detection via Spectral Analysis • Comparison of Normal vs. Faulty Patterns • Reporting Standards for Vibration Data	L & Ds	PPT, PDF	20 min
(c)	• Maintenance Implications of Vibration Trends • Integration with Power Performance Checks • Limitations of Vibration Analysis • Future Trends in EHM Technology • Summary of Key Concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how vibration data correlates with power performance degradation.
(b) Describe the role of spectral analysis in identifying specific engine faults.

12. Home Assignment and Follow Up	Prepare a brief note on how the techniques learned in this lesson will be applied in the next lesson plan regarding advanced diagnostic tools.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min